

(18CS138) IMAGE PROCESSING

(Professional Elective V)

COURSE OUTCOMES

At the end of the course, the students will develop ability to

1. Acquire basics on digital image processing system and various formats of digital images
2. Explain various geometrical and radiometric errors and their correction procedures, and pre-processing methods for image.
3. Experiment image processing tools on various images, develop practical knowledge and skills about pattern recognition tools
4. Illustrate necessary knowledge to design and implement a prototype of an image processing and pattern recognition application.
5. Identify difficulties in current existing image and pattern recognition and propose new methods for it.

UNIT I

The Digitized Image and Its Properties: Applications of image processing, image function, image representation, sampling, quantization, color images.

UNIT II

Metrics: Metrics and topological properties of digital images, histograms, image quality, noise image.

UNIT III

Image Preprocessing: Pixel brightness transformation, position dependent brightness correction, gray scale transformation; geometric transformation, local pre-processing image smoothing, edge detectors, zero-crossing, scale in image processing, canny edge detection, parametric edge models, edges in multi spectral images.

UNIT IV

Image Segmentation: Threshold detection methods, optimal thresholding, multispectral thresholding, thresholding in hierarchical data structures;

UNIT V

Image Segmentation: edge based image segmentation- edge image thresholding, edge relaxation, border tracing, border detection.

UNIT VI

Mathematical Morphology: Basic morphological concepts, four morphological principles, binary dilation, erosion, Hit or miss transformation, opening and closing; thinning and skeleton algorithms; Morphological segmentation -particles segmentation and watersheds, particle segmentation.

TEXT BOOKS

1. Millan sonka, Vaclav Hiavac, Roger Boyle, “Image Processing Analysis and Machine Vision”, 3rd Edition, CL Engineering , 2013.
2. Rafel C. Gonzalez, Richard E. Woods, “Digital Image Processing”, 3rd Edition, Pearson Education, 2008.

REFERENCE BOOKS

1. Julius T. Tou , Rafel C. Gonzalez, Addison, “Pattern Recognition Principles”, 1st Edition, Wesley publishing company.
2. Earl Gose, Richard Johnsonbaugh, “Pattern Recognition and Image Analysis”, 1st Edition, Prentice Hall of India Private limited, 2009.

(18CS140) HUMAN COMPUTER INTERACTION

(Professional Elective VI)

COURSE OUTCOMES

At the end of the course the student will be able to

1. Explain the human components functions regarding interaction with computer
2. Define key terms used in interaction design
3. Demonstrate Understanding of Interaction between the human and computer components
4. Implement Interaction design basics and apply Design rules
5. Produce Implementation supports, use Evaluation techniques

UNIT I

Introduction: Importance of user Interface – definition, importance of good design. Benefits of good design. A brief history of Screen design.

The graphical user interface: Popularity of graphics, the concept of direct manipulation, graphical system, Characteristics, Web user – Interface popularity, characteristics- Principles of user interface.

UNIT II

Design process: Human interaction with computers, importance of human characteristics human consideration, Human interaction speeds, understanding business junctions.

UNIT III

Methods for evaluation of interfaces with users: goals of evaluation, approaches, ethics, introspection, extracting the conceptual model, direct observation, constructive interaction, interviews and questionnaires, continuous evaluation via user feedback and field studies, choosing an evaluation method

UNIT IV

Screen Designing: Design goals – Screen planning and purpose, organizing screen elements, ordering of screen data and content – screen navigation and flow – Visually pleasing composition – amount of information – focus and emphasis – presentation information simply and meaningfully – information retrieval on web – statistical graphics – Technological consideration in interface design.

UNIT V

Windows: New and Navigation schemes selection of window, selection of devices based and screen based controls. Components – text and messages, Icons and increases – Multimedia, colors, uses problems, choosing colors.

UNIT VI

Software Tools: Specification methods, interface – Building Tools. Interaction Devices – Keyboard and function keys – pointing devices – speech recognition digitization and generation – image and video displays – drivers.

TEXT BOOKS

1. The essential guide to user interface design, Wilbert O Galitz, Wiley DreamaTech, 2007
2. Designing the user interface. 3rd Edition Ben Shneidermann , Pearson Education Asia, 2008.

REFERENCE BOOKS

1. Human – Computer Interaction. Alan Dix, Janet Fincay, Gre Goryd, Abowd, Russell Bealg, Pearson, 2005
2. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech, 2011.
3. User Interface Design, Soren Lauesen , Pearson Education, 2005.

WEB LINKS:

1. <http://www.scis.nova.edu/nova/hci/notes.html>
2. <http://courses.iicm.tugraz.at/hci/hci.pdf>
3. <http://www.ida.liu.se/~miker/hci/course.html>
4. <http://www.hcibook.com/>